5. Cystic fibrosis is an example of an inherited disease. A carrier of cystic fibrosis has one unaffected allele (F) and one affected allele (f).

(a) Two parents are planning to have a child. The father is a carrier of cystic fibrosis (Ff), the mother is unaffected.

(i) Circle the genotype of the unaffected mother. [1]

FF ff Ff

(ii) Complete the Punnett square to show the possible genotypes of the children. [2]

	F	f

(iii) State the chance of a child from these parents being born with cystic fibrosis. [1]

....

(iv) The possible genotypes of children from two carrier parents are shown below.

	F	f
F	FF	Ff
f	Ff	ff

These carrier parents thought they would have the same chance of having a child with cystic fibrosis as the parents in (a)(ii). Explain whether you agree. [1]

....

....

(b) Cystic fibrosis causes a build-up of thick mucus in the lungs. Mucus in the lungs helps bacteria to grow.

Underline **two** statements below which show ways white blood cells help to defend the body against bacterial infection. [2]

Carry oxygen to the bacteria

Ingest bacteria

Produce antibodies

Make glucose

(c) (i) Complete the table below to show how the number of bacteria doubles every 15 minutes. [2]

Time (minutes)	0	15	30	45	60
Number of bacteria	1	2	....	8	....

(ii) Antibiotic resistance is a concern. The list below describes the stages involved in the development of bacteria with antibiotic resistance. It is not in the correct order.

1. Many more bacteria survive antibiotic treatment.
2. There was a mutation to a gene in a few bacteria.
3. The survivors reproduce, passing on the mutated gene.
4. Bacteria with the mutation survive antibiotics.

Arrange the stages in the correct order starting with stage 2. [2]

2 ⟶ .... ⟶ .... ⟶ ....

(iii) Complete the following sentence by underlining the correct word in brackets. [1]

The development of antibiotic resistant bacteria is an example of **(evolution / vaccination / digestion)**.

12